# COMMERCIAL ITEM DESCRIPTION

## CLEANING COMPOUND, SEPTIC TANK (ENZYME BACTERIA TYPE FOR DRAINS)

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification P-C-440.

This description covers an enzyme-bacteria cleaning compound for use in septic tanks and for maintenance cleaning of drains.

Salient characteristics:

The product shall be a free-flowing powdered or granular material consisting of active enzymes combined with bacterial culture materials.

Starch liquefying ability. The product shall have a flow-through time of no more than 11.5 seconds when tested in the following manner:

Reagent. Prepare a buffered starch substrate by mixing 2 percent by weight sodium bicarbonate ($NaHCO_3$) and 1 percent by weight disodium phosphate ($Na_2HPO_4$) with gelatinized paste.[1/] This should be done by admixing the dry powders, which are then stored in a wide mouth screwcap bottle.

Equipment. An electric mixer with rheostat control, a strainer with 1.5 - 2.5 mm mesh, similar to a tea strainer, a 58° short-stemed fast filtering funnel, having a top I.D. of 7.5 mm, an orifice diameter of 4 mm, and a stem length of approximately 75 mm, and a stop watch.

Procedure. Weight out 1.00 g of product into a 250 ml beaker set in a water bath maintained at 58° to 60°C. Add 100 ml of 60° ± 0.5°C distilled water and stir slowly for 5 minutes. Increase the stirrer to about 1700 rpm and dump in 20 g of reagent all at once. Reduce the speed of stirring as necessary to avoid splashing, and stir for 5 minutes. Set up the funnel and strainer in the sink, and preheat by running hot water (50° - 75°C) through. The strainer should be nesting in the cone of the funnel to keep lumps of starch from plugging the funnel. Before pouring, turn off the hot water used for preheating and let it drain out of the funnel. The time measured is from the start of pouring until the last of the mixture leaves the cone of the funnel.

Protein digesting test. The product shall contain no less than 15,000 proteolytic/casein units per gram (P/C units/g) when tested in the following manner:

Reagents:

Acid-alcohol solution. Mix 25 ml glacial acetic acid, 225 ml of distilled water, 350 ml ethanol.

pH 8.0 buffer. Dissolve 15.4 g disodium phosphate, 15.0 g sodium citrate, and 10.3 g boric acid in 500 ml of distilled water. Adjust the pH to 8.0 with dilute sodium hydroxide.

Substrate. Prepare a 2 percent stock solution of casein by dissolving 5.0 g casein in 200 ml of buffer solution. Heat in water bath at 90°C with occasional stirring until casein dissolves, cool to 40°C and bring to 250 ml total. This solution will keep 5 days under refrigeration but must be reheated to redissolve before diluting to 0.2 percent by pipetting 10 ml into a 100 ml volumetric flask and making the solution up to the mark.

Enzyme standard. Adjust a bacterial enzyme of known potency to 15,000 p/c units per gram by mixing the material with table salt.[2/] The prepared standard will keep for six months if stored in the freezer.

Enzyme extracts. Extract 1.0 g of enzyme standard or material to be tested in 100 ml of water in a 250 ml beaker. Place the beakers in a 40.0° ± 0.5°C water bath for 35 to 40 minutes, with occasional stirring. Filter 15 ml of the extract and place the filtrate in the refrigerator at 4°C. The standard enzyme should be extracted fresh each day of testing and kept in the refrigerator.



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Procedure. Pipette 10 ml of substrate into each of six 13 by 122 mm test tubes. Chill in the refrigerator for 15 minutes and pipette 1.0 ml of enzyme standard extract into three of them and 1.0 ml of test material extract into the other three. Mix well and place the tubes at once into the $40.0^{\circ} \pm 0.5^{\circ}C$ water bath and digest. At the end of 30, 35 and 40 minutes remove a pair of tubes, one containing test material extract and one containing enzyme standard extract, and pipette 1.0 ml of acid-alcohol solution into each tube. Agitate the tubes to ensure thorough mixing and replace in the bath for 1-2 minutes before making a visual comparison. Shake the closed tube vigorously before comparing. The tubes containing test material extract should have equal or less turbidity than those containing standard enzyme extract.

Anaerobic bacteria test. The product shall develop 1.5 billion colonies per gram when tested in the following manner:

Weigh out 1.00 g of product into 50.0 ml of sterile solution blank containing glass beads. Restopper and allow to stand for 5 minutes, then agitate vigorously for 5 additional minutes. Dilute the sample serially 1 to 5,000,000, using serological pipettes, and 0.50 ml of the final dilution is added to 24.5 ml of Brewer's Anaerobic Agar, which has been previously liquefied and cooled to $45^{\circ}C$ in a cotton stoppered 50 ml Erlenmeyer flask. The agar is swirled carefully to mix the 0.50 ml of sample with it while avoiding entrainment of air. Pour the agar into an oval culture tube, 23 x 11.5 mm cross section, 152 mm length, fill it to a depth of about 5 inches. Immediately overlay the agar with 18-20 mm of sterile vaseline, paraffin or other inert sterile ingredient. The sterile sealant should be less than $55^{\circ}C$ at the time of pouring, and as close to $50^{\circ}C$ as possible. Within several minutes the liquefied agar and sample will turn yellow, indicating that a reduced medium has been obtained. Then incubate the tube at $36^{\circ} \pm 1^{\circ}C$ for 24 hours. Mark off the tube into 1-inch sections, count the colonies in three sections, and average them. The average count should be not less than 30 colonies to the inch, equivalent to 1.5 billion colonies per gram.

pH. The product shall have a pH in the range of 6.0-8.0, when 10 g of the product is dissolved in 60 ml of distilled water at room temperature and tested in accordance with ASTM E 70 Method.

Salmonella test. The product shall be free of Salmonella when tested in accordance with section 4.5(s) of the USDA's Microbiology Laboratory Guidebook.

The issue of ASTM E 70 in effect on the date of the solicitation shall be used to determine compliance with this requirement.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices, and is the same product sold in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements. The manufacturer shall utilize recovered materials to the maximum extent practicable. The product shall comply with 16 CFR-1500 Federal Hazardous Substances Act and Regulation promulgated thereunder.

Packaging, packing, and marking. Packaging, packing, and marking shall be as specified in the contract or order.

Notes: The product is intended for the regular maintenance of drains, traps, pipelines and septic tanks. It is not intended for emergency cleaning of plugged drains, etc. The product should be used in accordance with the manufacturers instructions.

ASTM standards are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

The Microbiology Laboratory Guidebook is available from the US Department of Agriculture, Consumer and Marketing Service, Technical Services Division, Washington, DC 20036.

1/ Wheat Paste is available from the Arthur Hoyt Co., PO Box 24, Hickville, NY 11802.
2/ Bacterial Enzyme standard is available from the George A. Jeffries, Co., PO Box 709, Salem, VA 24153.

MILITARY INTEREST:

NO INTEREST

CIVIL AGENCY COORDINATING ACTIVITIES:

AGR - AFS, CMS
INT - BLM
GSA - FSS
VA - OSS

PREPARING ACTIVITY:

GSA - FSS



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